

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Database Management System (New)

Semester: Fall

Year : 2025
Full Marks : 100
Pass Marks : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Explain the three levels of data abstraction in DBMS with suitable examples. Also define data independence and why it is important. 7
- b) Design an ER Diagram for a "Hospital Management System." Your diagram must include: 8
 - i. At least four entity sets (e.g., Patient, Doctor, Room, Medicine).
 - ii. Examples of multi-valued and derived attributes.
 - iii. Clearly marked participation constraints and cardinalities.
2. a) Given the relations: 7

Student(SID, Name, Major) . . .
Course (CID, CName, Credits)
Enroll(SID, CID, Grade)

Write relational algebra expressions for the following:

 - i. Find the names of students who have a grade of 'A'.
 - ii. List all SIDs of students who have not enrolled in any course.
 - iii. Perform a Natural Join between the two tables.
 - iv. Find the names of students enrolled in 'CS101'.
- b) Consider the relational database 8

Employees (EmpID, Name, DeptID, Salary, JoinDate)
Departments (DeptID, DeptName, Location)
Projects (ProjectID, ProjectName, Budget, DeptID)

Write SQL queries for:

 - i. Create the table Employees with appropriate constraints.
 - ii. Find the Name, Salary of employee of 'IT' department.

- iii. Increase the salary of employees in the "IT" department by 10%.
 - iv. Delete all projects with a budget less than 40,000.
3. a) Explain about different types of constraints in Database? Explain each with suitable example. 7
- b) Explain about different anomalies we face before normalization? Also explain about 2NF, 3NF, BCNF with suitable example. 8

OR

What is closure of attribute sets? How is it computed? For $R = (A, B, C, D, E, F)$ and $F = \{A \rightarrow BC, E \rightarrow CF, B \rightarrow E, CD \rightarrow EF\}$, Compute $(AB)^+$.

4. a) What is access control in a database system? Explain different access control mechanisms used in DBMS. 7
- b) What is Query Processing? Explain the steps involved in query processing with suitable diagram. 8
5. a) Describe different file organization techniques used in DBMS. Explain heap file organization with its merits and demerits. 7
- b) What is Conflict Serializability? Explain how the Two-Phase Locking (2PL) protocol ensures that a schedule is serializable. 8

OR

Explain Lock Based Protocol along with its pitfalls. Let we have two accounts A and B with values 5000 and 10000 respectively. Transaction T1 transfers 1000 from A to B. Transaction T2 display sum of A and B. Apply appropriate Locks in above transaction T1 and T2 and make serial schedule.

6. a) Describe the Log-Based Recovery mechanism. Differentiate between Deferred Database Modification and Immediate Database Modification techniques. 7
- b) What is Object Relational Mapping (ORM)? Explain its role in modern application development. 8
7. Write short notes on: (Any two) 2×5
- a) NoSQL Databases and its characteristics
 - b) Transaction State Diagram
 - c) Stored Procedure